

CBSE Class 12

Physics

Previous Year Question Paper 2016

Series: ONS

Code no. 55/1/C

- Please check that this question paper contains **15** printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **26** questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minutes time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer script during this period.

PHYSICS (Theory)

Time Allowed: **3** hours

Maximum Marks: **70**

General Instructions:

1. **All** questions are compulsory. There are **26** questions in all.
2. The question paper has **five** sections: Section A, Section B, Section C, Section D and Section E.
3. Section A contains **five** questions of **one** mark each, Section B contains **five** questions of **two** marks each, Section C contains **twelve** questions of **three** marks each, Section D contains **one** value based question of **four** marks each and Section E contains **three** questions of **five** marks each.
4. There is no overall choice. However there is an internal choice that has been provided in **one** question of two marks and one question of **three**

marks and all three questions of **five** marks weightage. You have to attempt only **one** of the choices in such questions.

5. You may use the following values of physical constants wherever necessary:

$$c=3\times 10^8 \text{ m/s}$$

$$h=6.63\times 10^{-34} \text{ Js}$$

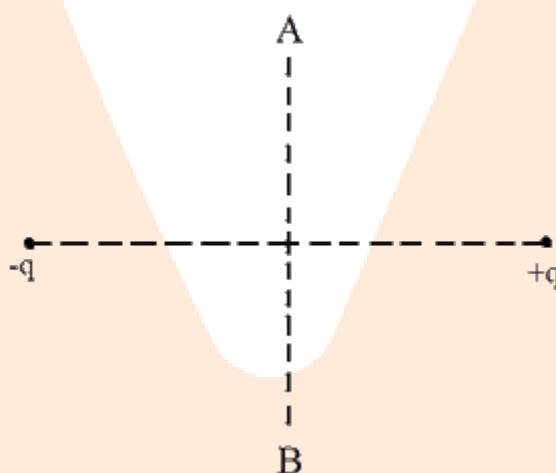
$$e=1.6\times 10^{-19} \text{ C}$$

$$\mu_0=4\pi\times 10^{-7} \text{ T m A}^{-1}$$

$$\frac{1}{4\pi\epsilon_0}=9\times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

PART-A

1. A charge 'q' is moved from a point A above a dipole moment 'p' to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process. 1 Mark



Ans: Since the potential at a point perpendicular to a dipole is zero or when it is placed in an equatorial position with respect to the dipole it will have zero potential and hence the work done on moving the charge will be zero as:

$$W=qV_{AB}=q(V_B-V_A)$$

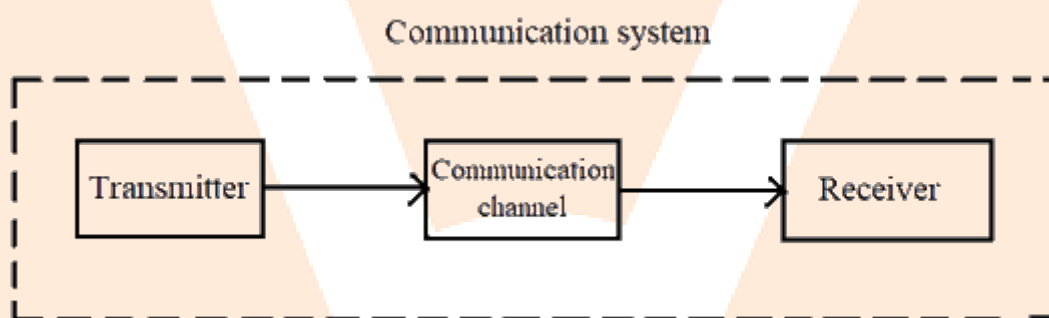
$$\Rightarrow W=q(0)=0$$

2. In what way is the behavior of a diamagnetic material different from that of a paramagnetic, when kept in an external electric field? 1 Mark

Ans: The property that differentiates diamagnetic and paramagnetic substances is a physical quantity known as magnetic susceptibility. Magnetic susceptibility measures the ability of the substance to execute magnetization when placed in a magnetic field. This value is positive in the case of paramagnetic materials and negative in the case of diamagnetic materials.

3. Name the essential components of a communication system. 1 Mark

Ans: The basic components of a communication system are Transmitter, communication channel and a receiver. These are arranged as shown in the block diagram given below:



The essential components of a communication system as given in the diagram is a transmitter which converts the message signal into a suitable form for transmission, a communication channel which acts as a medium for propagation of the signal and a receiver which it recovers back the original signal at the receiving end.

4. Why does sun appear to be red at sunrise and sunset? 1 Mark

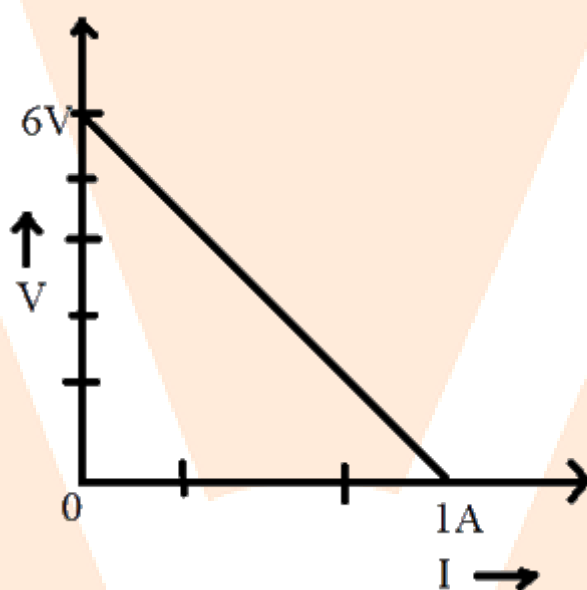
Ans: The colours that are formed in nature are explained by the Rayleigh's scattering of light which says that the intensity of light is inversely proportional to the fourth power of its wavelength.

$$\Rightarrow I = \frac{1}{\lambda^4}$$

We know that the red colour of the visible light spectrum has a greater wavelength. Therefore, according to Rayleigh's scattering of light since red has

a large wavelength its scattering power is less. Hence, the sun appears to be red at sunset and sunrise as a result of least scattering of red light due to long wavelength range.

5. The plot variation of the potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell? **1 Mark**



Ans: Emf of a cell is said to be the potential difference when no current flows through the circuit or when zero current is drawn from the cell. Hence, from the graph when current is 0 then the voltage or the emf is seen to be 6V.

Thus the emf, $\epsilon = 6V$.

The potential difference in terms of internal resistance of a cell is given as:

$$V = \epsilon - Ir \text{ -----(1)}$$

From the graph when the voltage is zero it can be seen that the current is 1A. Hence by substituting in the above equation (1) we get:

$$0 = 6 - (1)r$$

Hence, the internal resistance is, $r = 6\Omega$. This is the internal resistance of the combination of the cells and since there are three resistors the internal resistance of each cell is hence 2Ω .

PART-B

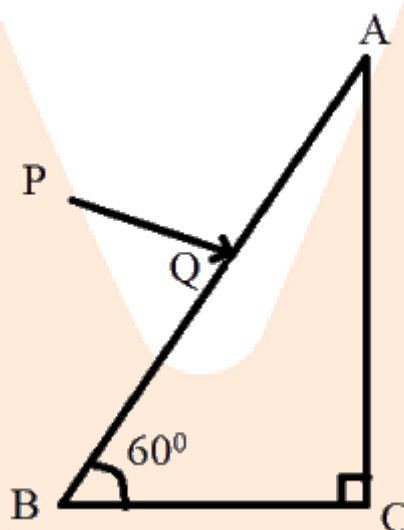
6. Define modulation index. Why is it kept low? What is the role of a bandpass? 2 Marks

Ans: Modulation index also known as modulation factor is defined as the ratio of the change in the amplitude of the carrier wave to that of the modulating signal or the original signal. It signifies the extent to which amplitude of the carrier wave is changed due to the mixing of the modulated wave.

It is kept low in order to avoid distortion of the signal because when a signal is over-modulated then the negative peak of the time varying signal is lost producing a discontinuous wave and hence the modulation index is kept low.

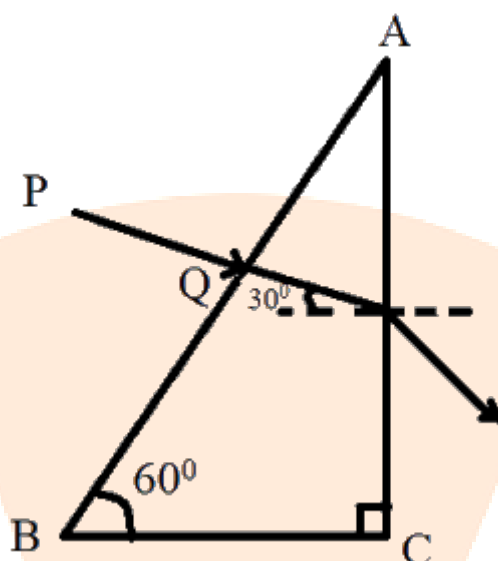
The role of a bandpass is to reject frequencies that are high or low and in turn allows a certain band of frequencies to pass through it.

7. A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge ? Justify your answer. 2 Marks



Ans: Since the light is incident normally on the surface AB of the triangle there will not be any bending of light ray until it reaches the face AC. At face AC since the light needs to travel from a denser to a rarer medium refraction takes place and the light bends away from the normal at face AC.

The ray path will hence be as follows:



By applying snell's law we have:

$$\mu = \frac{\sin r}{\sin i}$$

The refractive index is given to be 1.5 we get:

$$\Rightarrow \sin i = \frac{1}{1.5} = \frac{2}{3}$$

$$\Rightarrow i = \sin^{-1}\left(\frac{2}{3}\right) \approx 41.8^\circ$$

Here, since $\angle i$ is greater than 30° . Hence, AC will be the emerging face of the prism.

8. Calculate the de-Broglie wavelength of the electron orbiting in the $n=2$ state of the Hydrogen atom. 2 Marks

Ans: The ionization potential is given as the accelerating potential given to the electron for it to gain energy and jump to its excited state and this is also the kinetic energy gained by it to move to its excited state. This state is given as, $n=2$. The equation for kinetic energy is thus given as:

$$K.E = \frac{13.6Z^2}{n^2}$$

The atomic number of hydrogen is said to be 1, that is, $Z=1$. Hence the given values are substituted to get:

$$K.E = \frac{13.6(1)^2}{(2)^2} = \frac{13.6}{4} = 3.4\text{eV}$$

Since, the charge of an electron is $e=1.6\times 10^{-19}\text{C}$. Hence the kinetic energy in terms of volts is:

$$K.E = 3.4 \times 1.6 \times 10^{-19}\text{V}$$

The de-Broglie wavelength is given as:

$$\lambda = \frac{h}{p}$$

However, the value of $p = \sqrt{2mK}$

Hence, by substituting in the above equation for de-Broglie wavelength we have:

$$\lambda = \frac{h}{\sqrt{2mK}}$$

We know that, Planck's constant, $h=6.63\times 10^{-34}\text{Js}$ and the mass of an electron is, $m=9.1\times 10^{-31}\text{kg}$. Thus, by substituting all the given values we get:

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 3.4 \times 1.6 \times 10^{-19}}}$$

On further simplification we have:

$$\Rightarrow \lambda \approx 0.67\text{nm}$$

9. Define ionization energy.

How would the ionization energy change when an electron in hydrogen atom is replaced by a particle of mass 200 times that of the electron but having the same charge?
2 Marks

Ans: Ionization energy is defined as the energy that is required to pluck out or remove an electron from its valence shell, that is, its outermost shell of the atom. It is also said to be the energy required for the electron to jump from its ground state to its excited state.

The ionization energy of an electron is given by the following equation:

$$E_0 = \frac{2\pi^2 m k^2 Z^2 e^4}{n^2 h^2}$$

From this equation it can be seen that all the variables excluding the mass of the electron are constants and hence the following relation is established:

$$E_0 \propto m$$

Since the ionization energy is directly proportional to the mass of the electron hence with an increase in the mass by 200 times the ionization energy also increases by the same amount.

Or

Calculate the shortest wavelength of the spectral lines emitted in Balmer series. [Given Rydberg's constant, $R=10^7 \text{ m}^{-1}$]

Ans: The wavelengths lying in the range of the Balmer series is given by the equation:

$$\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Where, n_1 and n_2 are the transition states of the electron in the spectral series respectively. Here, n_2 can take the values, $n_2=3,4,5,\dots,\infty$ and $n_1=2$. For the shortest wavelength with the least energy n_2 takes the value ∞ . The Rydberg's constant is already given to be, $R=10^7 \text{ m}^{-1}$. Hence, by substituting all these values we get the wavelength to get:

$$\frac{1}{\lambda} = 10^7 \left[\frac{1}{(2)^2} - \frac{1}{\infty} \right]$$

On further simplification we get:

$$\frac{1}{\lambda} = 10^7 \left[\frac{1}{(2)^2} - 0 \right] = \frac{10^7}{4}$$

$$\Rightarrow \frac{1}{\lambda} = 2.5 \times 10^6$$

On reciprocating we get:

$$\Rightarrow \lambda \approx 0.4 \times 10^{-6} = 4 \times 10^{-7} \text{ m}$$

Hence this is the shortest wavelength of the spectral lines emitted in the Balmer series.

10. A battery of emf 12V and internal resistance 2 ohm is connected to a 4 ohm resistor as shown in the figure. 2 Marks

(i) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.

Ans: The emf of the cell when no current flows through the circuit, the internal resistance, r , of the circuit and the resistance, R_{net} of the resistor connected in the circuit are given.

Given,

$$\varepsilon = 12\text{V}$$

$$R = 4\Omega$$

$$r = 2\Omega$$

By applying the ohm's law equation we get:

$$V = IR_{\text{net}}$$

$$\Rightarrow I = \frac{V}{R+r}$$

The effective or the equivalent resistance is with respect to the 4 ohm resistor since only a single resistor is present in the circuit. The total current through the circuit can hence be found out using this equation by substituting the given values:

$$I = \frac{12}{4+2} = \frac{12}{6} = 2\text{A}$$

$$\Rightarrow I = 2\text{A}$$

The potential difference shown by the voltmeter is given by the formula:

$$V = \varepsilon - Ir$$

By substituting the given values we have:

$$\Rightarrow V = 12 - 2 \times 2 = 12 - 4$$

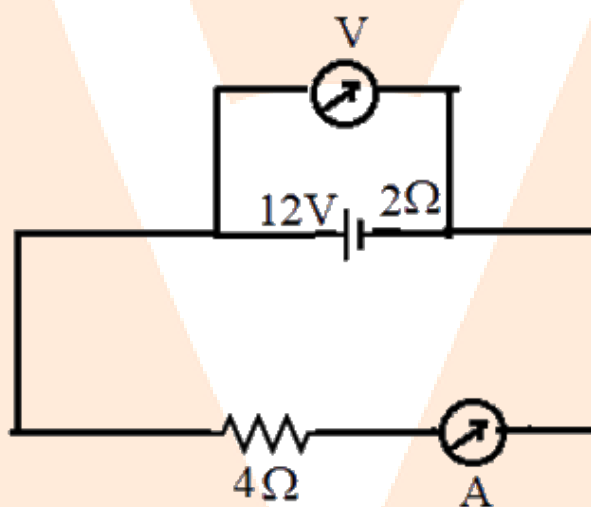
$$\Rightarrow V = 8V$$

This is the potential difference across the cell. The potential difference across the resistor is measured as the potential drop across the resistor and hence the voltage in the 4 ohm is obtained by applying the ohm's law equation again we get:

$$V = 2(4) = 8V$$

This is the potential in the resistor and hence it can be seen that both these voltages measured by the voltmeter are the same.

(ii) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit?



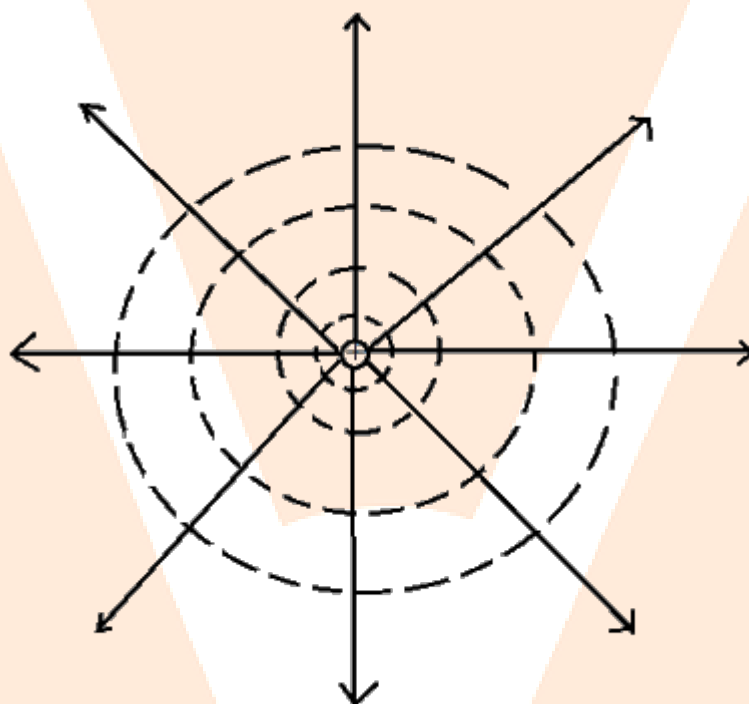
Ans: An ammeter is a device which measures current. An ammeter has a very low resistance when connected in series to the circuit element through which the current must be measured and thus this helps in maintaining the current in a circuit to be the same throughout so that the ammeter measures the current accurately. Likewise the voltmeter is connected in parallel to the device through which the voltage needs to be measured because it offers high resistance and hence the potential difference is obtained accurately.

PART-C

11. Define an equipotential surface. Draw equipotential surfaces: 3 Marks

(i) in the case of a single point charge and

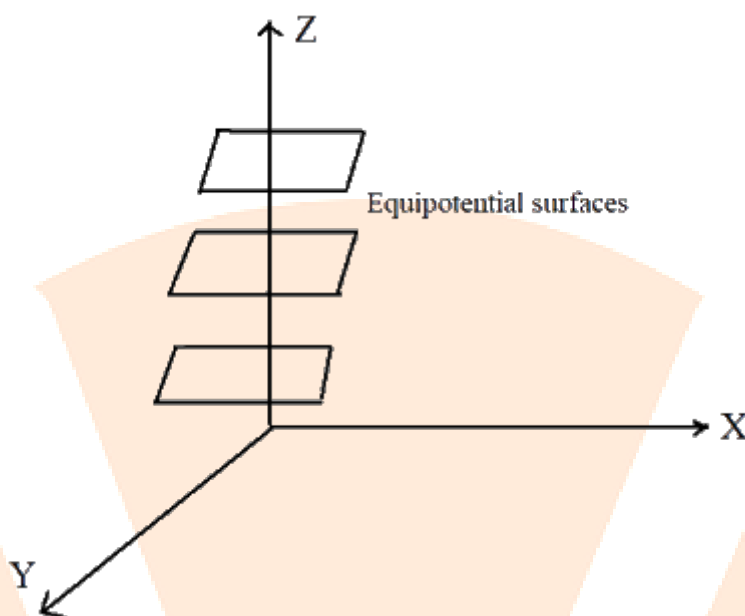
Ans: The equipotential surface for a single point charge is illustrated in the diagram below:



(ii) in a constant electric field in Z-direction.

Why the equipotential surfaces about a single charge are not equidistant?

Ans: The equipotential surface for a constant electric field in Z-direction is illustrated in the diagram below:



The electric field is given by the formula:

$$E = \frac{kq_1q_2}{r^2}$$

Hence, we can see that,

$$E \propto \frac{1}{r^2}$$

Since, the electric field is inversely related to the distance of separation, the equipotential surfaces for a single charge are not equidistant from each other as the electric field is not constant. Also, the potential is given as:

$$V = \frac{kq}{r^2}$$

$$\Rightarrow V \propto \frac{1}{r^2}$$

Thus, the potential is also inversely proportional to its distance and thus the formation of the equipotential surfaces are not equidistant.

(iii) Can electric field exist tangential to an equipotential surface? Give reason.

Ans: As we know, equipotential surfaces are ones that have the same or constant potential throughout or at all points in its surface and hence there will be no work done in moving a charge in presence of a field because the difference between the potentials at point A and another point B will be zero.

Thus, if the electric field is tangential to the surface then this component will mean that there is some work done which is in contrast to the principle of equipotential surfaces and hence electric field cannot exist tangential to the surface and is rather always perpendicular to the surface.

12.

3 Marks

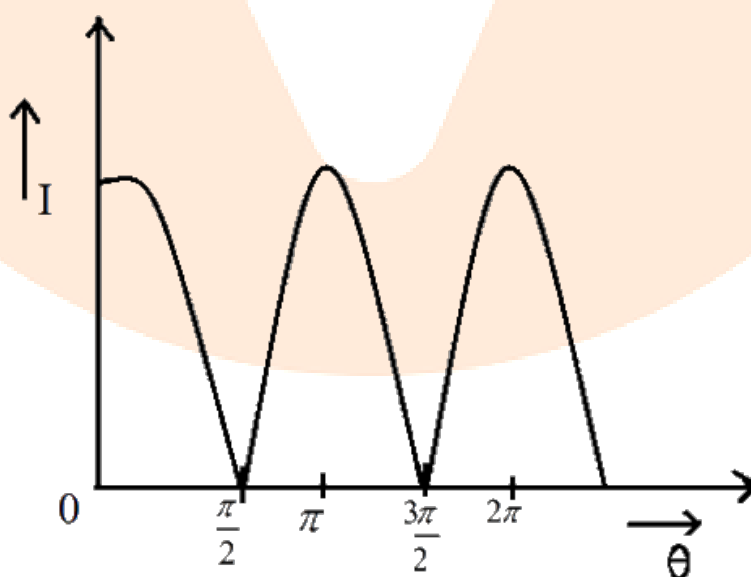
(i) State law of Malus.

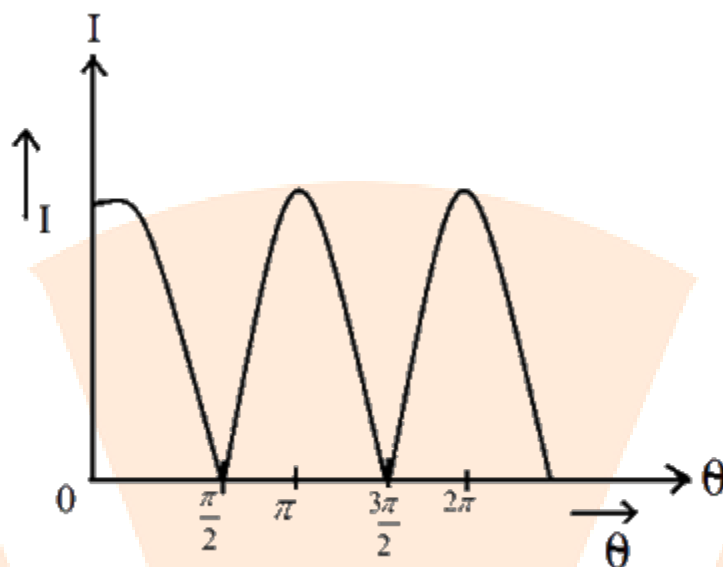
Ans: When a beam of completely plane polarized light is passed through an analyzer the intensity 'I' of transmitted light varies directly as the square of the cosine of the angle between the transmitted directions of polariser and analyzer. The equation of law of Malus is given as:

$$I = I_0 \cos^2 \phi$$

(ii) Draw a graph showing the variation of intensity (I) of polarized light transmitted by an analyzer with angle (θ) between polarizer and analyzer.

Ans: The graph showing the variation of the intensity of a polarized light by its polarizing angle is shown below:





(iii) What is the value of refractive index of a medium of polarizing angle 60° ?

Ans: The relation between refractive index and the polarizing angle is given as:

$$\mu = \tan i_p$$

Since, the polarizing angle is 60° we get the refractive index to be:

$$\mu = \tan(60^\circ) = \sqrt{3}$$

Hence, we get the refractive index to be:

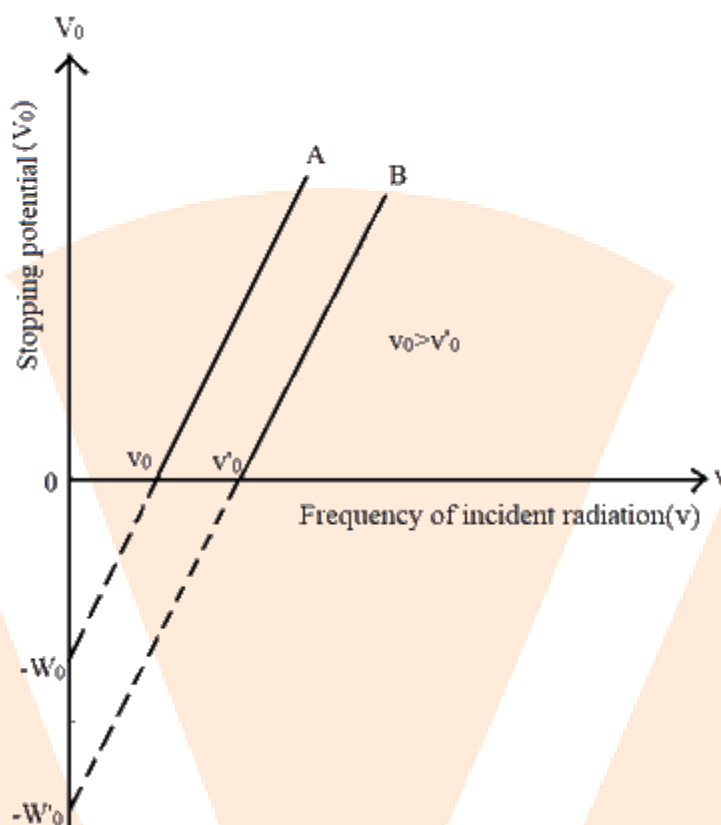
$$\mu = 1.73$$

Therefore, this is the refractive index for the given polarizing angle

13. Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials A and B having threshold frequencies $\nu_A > \nu_B$. 3 Marks

(i) In which case is the stopping potential more and why?

Ans: The variation for photoelectric current with anode potential or collector plate potential is shown in the graph below:



The stopping potential and frequency of incident light are directly proportional to each other. Also, with an increase in the frequency of incident light the kinetic energy of the ejected electrons also increases until the kinetic energy reaches its maximum.

The graph shows that the potential is gradually decreased until stopping potential is reached. Hence, it can be concluded that with an increase in frequency the retarding potential applied must be greater to obtain its stopping potential since the photons ejected will be having high kinetic energy.

Thus, the stopping potential for material A is more since its kinetic energy is more hence requires more potential for it to reach its stopping potential.

From the equation for kinetic energy in terms of its stopping potential is:

$$K = eV_0$$

In accordance to the Einstein's theory of photoelectric effect the kinetic energy is given as:

$$K = h\nu - h\nu_0$$

Hence by equating these equations we get:

$$eV_0 = h\nu - h\nu_0$$

On simplifying further we get:

$$eV_0 = h(\nu - \nu_0)$$

$$\Rightarrow V_0 = \frac{h}{e}(\nu - \nu_0) \text{ -----(1)}$$

(ii) Does the slope of the graph depend on the nature of the material used? Explain

Ans: No, the slope does not depend on the nature of the material. This can be seen from equation (1) which is in accordance to the straight line equation $y = mx + c$ and hence the slope of the graphs is given by the term $\frac{h}{e}$. Thus, the slopes of the graphs depend only upon the constants that is the Plank's constant and the charge of the ejected electrons and not on the nature of the material.

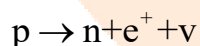
14.

3 Marks

(a) Write the basic nuclear process involved in the emission of β^+ in symbolic form, by a radioactive nucleus.

Ans: A β^+ decay is a spontaneous process where the nuclear process that takes place is the emission of a positron from the nucleus, that is, a proton turns or gets converted into a neutron inside the nucleus called as the positron.

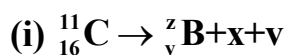
In a β^+ emission, the mass number remains constant but however the atomic number decreases by one due to the emission of the positron and the decay also leads to the creation of a new particle known as neutrino. Hence, the reaction is as follows:



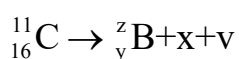
Here, e^+ and ν represent the positron and the neutrino respectively.

(b) In the reactions given below:

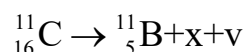
Find the values of x, y and z and a, b and c.



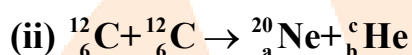
Ans: The given reaction is:



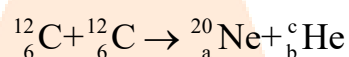
The mass number and atomic number of boron is already known which is 11 and 5 respectively in accordance to its values given by the periodic table of elements. Hence, the given equation becomes:



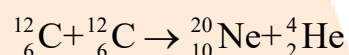
Here, x denotes the positron released in β^+ decay. Since the mass number remains unchanged while the atomic number decreases by 1, the above equation becomes:



Ans: The given reaction is:



The above reaction is an example of alpha decay where the Helium nucleus is said to be the alpha particle that is emitted during α -decay. Hence, the above equation becomes:



Here, He is the helium nucleus. Therefore, the values are:

$$y=1$$

$$z=0$$

$$a=10$$

$$b=2 \text{ and}$$

$$c=4$$

Also, x value is the positron β^+ .

15.

3 Marks

(i) Derive the expression for drift velocity for free electrons.

Ans: The velocity with which the electrons randomly move with inside a conductor is called as thermal velocity and this is the velocity of the free electrons when there is no electric field applied.

When considering the average thermal velocity of these electrons it was seen that they tend to move in different directions and hence it was difficult to determine their average velocity since their velocities would get cancelled out.

Hence the average thermal velocity of the electrons was given as:

$$v_{\text{avg}} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n} = 0 \quad \text{-----(1)}$$

When an external electric field is applied to the conductor then the electrons tend to accelerate towards the positive potential of the conductor but however they tend to bump into the fixed positively charged ions with a greater mass present inside the metal lattice and hence lose their momentum. Hence, the net velocity gained will be small known as the drift velocity.

Due to the applied electric field, the electrons tend to experience some force and this force is given as:

$$\vec{F} = -e\vec{E} \quad \text{-----(2)}$$

Where, e is the charge of the electron

From Newton's second law of motion we have the acceleration of the electron particles to be:

$$a = \frac{\vec{F}}{m_e}$$

By substituting equation (2) in the above equation we get:

$$a = -\frac{e\vec{E}}{m_e} \quad \text{-----(3)}$$

There will be a gain in velocity before it suffers the next collision and hence when it collides with the positive ion it loses all its energy and this process occurs for all the electrons present in the conductor. Thus,

Velocity of 1st electron:- $v_1 + a\tau_1$

Velocity of 2nd electron:- $v_2 + a\tau_2$ and so on for the other electrons

Hence, the drift velocity can be referred as the average speed gained by the free electrons in opposite direction of the externally applied electric field. Hence it will be given as:

$$v_d = \frac{(v_1 + a\tau_1) + (v_2 + a\tau_2) + (v_3 + a\tau_3) + \dots + (v_n + a\tau_n)}{n}$$

This can be written as:

$$v_d = \frac{(v_1 + v_2 + v_3 + \dots + v_n)}{n} + \frac{(a\tau_1 + a\tau_2 + a\tau_3 + \dots + a\tau_n)}{n}$$

From equation (1) it can be seen that the average thermal velocity is 0. Hence, we have:

$$v_d = 0 + \frac{a(\tau_1 + \tau_2 + \tau_3 + \dots + \tau_n)}{n}$$

Let the average mean time of all the electrons be τ and hence we have:

$$v_d = a\tau$$

$$\text{Where, } \tau = \frac{(\tau_1 + \tau_2 + \tau_3 + \dots + \tau_n)}{n}$$

From equation (3) we have:

$$\vec{v}_d = \frac{e\vec{E}\tau}{m_e}$$

This is the expression for drift velocity.

(ii) How does drift velocity for electrons in a metallic conductor vary with increase in temperature? Explain.

Ans: The drift velocity of the free electrons in the conductor are inversely related to the temperature because when temperature is increased then they gain more

energy to move faster and thus collide into more number of positive ions decreasing its drift velocity.

16.

3 Marks

(i) When an AC source is connected to an ideal inductor show that the average power supplied by the source over a complete cycle is zero.

Ans: The alternating source that is connected to the inductor produces a current correspondingly which tends to lag behind the voltage of the source by $\frac{\pi}{2}$ radian because this is the phase difference between current and voltage in an inductor. Thus, the instantaneous values of emf and current will be:

$$V = V_0 \sin \omega t \text{ -----(1)}$$

$$I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$$

The equation for current can be rewritten as:

$$I = -I_0 \sin \left(\frac{\pi}{2} - \omega t \right) \quad [\text{Since, } \sin(-\theta) = -\sin(\theta)]$$

By applying the basic trigonometry formula we have:

$$\text{Since, } \sin \left(\frac{\pi}{2} - \theta \right) = \cos(\theta) \text{ we have:}$$

$$I = -I_0 \cos \omega t \text{ -----(2)}$$

The small amount of work done in a small interval of time dt is given as:

$$dW = P dt \text{ -----(3)}$$

The power in terms of voltage and current is given as:

$$P = VI$$

By substituting equations (1) and (2) in the above equation we get:

$$P = -V_0 \sin \omega t \times I_0 \cos \omega t$$

$$\Rightarrow P = -I_0 V_0 \sin \omega t \cos \omega t$$

This value of power is substituted in equation (3) to get:

$$dW = -I_0 V_0 \sin \omega t \cos \omega t \, dt \quad \text{-----(4)}$$

Hence, the average power dissipated over a cycle of an inductor is given as:

$$P = \frac{W}{t}$$

To find the total power over this interval of time we have:

$$P = \frac{1}{t} \int_0^T dW$$

By substituting equation (4) we get:

$$P = \frac{1}{t} \int_0^T -I_0 V_0 \sin \omega t \cos \omega t \, dt$$

By taking out the common terms we get:

$$P = -\frac{I_0 V_0}{t} \int_0^T \sin \omega t \cos \omega t \, dt$$

$$\Rightarrow P = -\frac{I_0 V_0}{2t} \int_0^T 2 \sin \omega t \cos \omega t \, dt$$

Since, $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\Rightarrow P = -\frac{I_0 V_0}{2t} \int_0^T \sin 2\omega t \, dt$$

On integrating we get:

$$\Rightarrow P = -\frac{I_0 V_0}{2t} \left[\frac{\cos 2\omega t}{2} \right]_0^T$$

We know that, $T = 2\pi\omega$

$$\Rightarrow \omega = \frac{2\pi}{T}$$

Hence we substitute back to get:

$$\Rightarrow P = -\frac{I_0 V_0}{2t} \left[\frac{\cos \frac{4\pi}{T} t}{2} \right]_0^T$$

$$\Rightarrow P = -\frac{I_0 V_0}{4t} \left[\cos \frac{4\pi}{T} t \right]_0^T$$

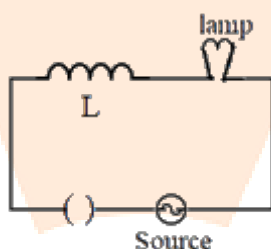
On further simplification we get:

$$\Rightarrow P = -\frac{I_0 V_0}{4t} [\cos 4\pi - \cos 0] = -\frac{I_0 V_0}{4t} [1 - 1]$$

$$\Rightarrow P_{\text{avg}} = 0$$

Hence proved that the average power of an ideal inductor over a single cycle is zero.

(ii) A lamp connected in series with an inductor and an AC source. What happens to brightness of the lamp when the key is plugged in and an iron rod is inserted inside the inductor?



Ans: The brightness of the lamp after the key is plugged in, decreases. The reason for this decrease in brightness is as follows.

When the iron rod is inserted into the inductor, current is produced in the conductor which in turn generates an induced emf. Hence, the inductance increases. The reactance of the capacitor in the given circuit is given by the equation:

$$X_L = \omega L \text{ -----(1)}$$

Where,

L is the inductance of the inductor

ω is the angular frequency of the capacitor

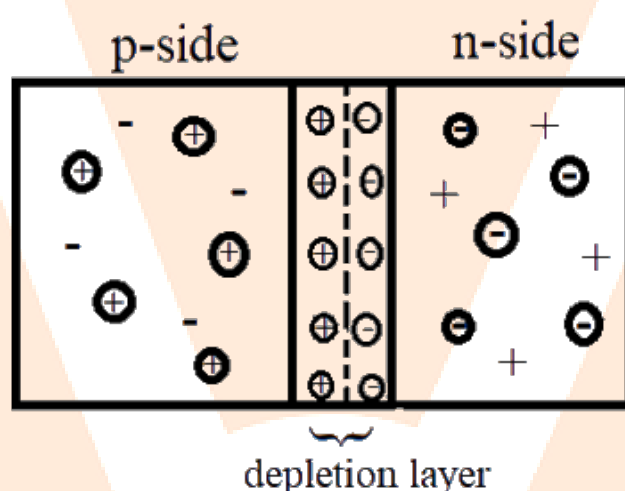
From equation (1) it can be seen that the inductance and the reactance of an inductor are directly proportional to each other. Hence, when the inductance value is increased the reactance will be more, which implies that the current flowing through the lamp will also decrease because reactance is the resistance offered by the inductor. Therefore, the brightness of the lamp tends to reduce.

17.

3 Marks

(i) Explain with the help of a diagram the formation of a depletion region and barrier potential in a pn junction

Ans: When the holes and electrons are recombined then a layer which is free of any free charges is formed called the depletion layer. The diagram for pn junction is as shown below:

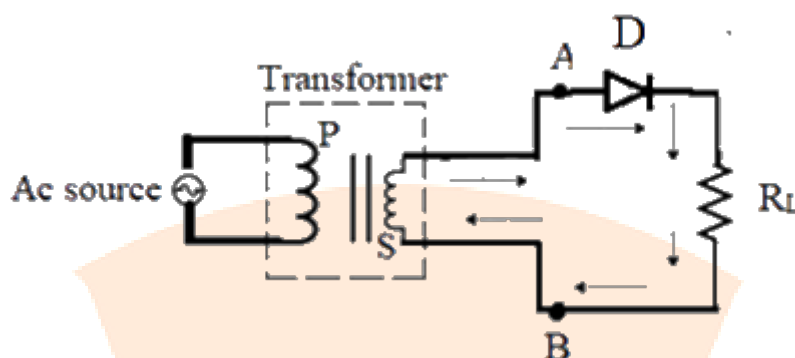


The width of the depletion layer will decrease in its size when forward biased. and vice-versa during the reverse biasing. In forward biasing, since the voltages of the terminal and the potential produced in the p-n junction will be in opposite directions to each other and recombination is less, making the depletion layer less. Since the recombinations will be more when reverse biased the depletion layer will increase in this case.

Potential barrier: The potential barrier is formed due to the difference in voltages between the terminal potential difference and the potential produced in the pn-junction diode due to the flow of charge carriers.

(ii) Explain the working of a half-wave rectifier.

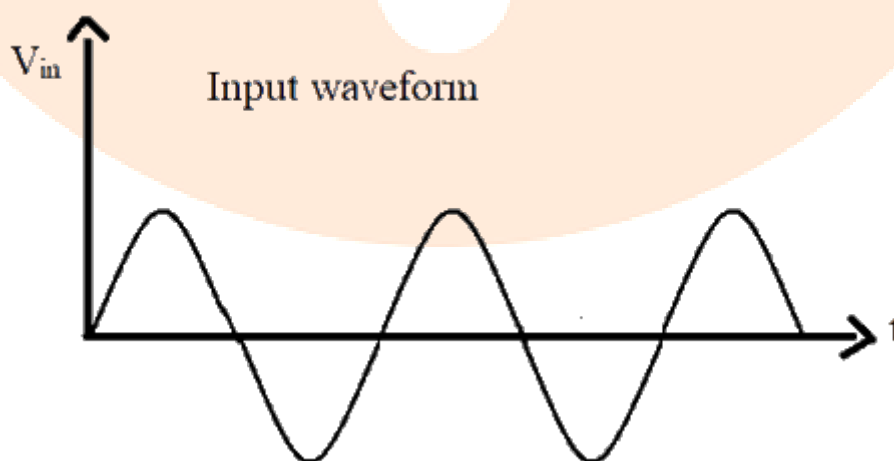
Ans: The diagram for the half-wave rectifier is given below:

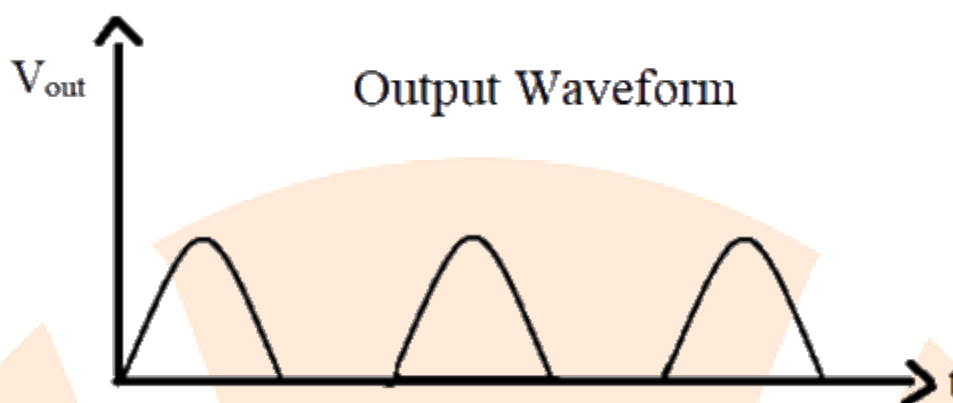


The half wave rectifier as the name suggests produces only half of the output waveform produced by the rectifier only during its positive half cycle. The working principle of the half wave rectifier is as follows. The ac source produces an alternating current which the transformer induces and the current starts to flow through the circuit. During the positive cycle the point A takes a positive polarity while point B takes a negative polarity making the diode forward biased.

Thus, current flows through the circuit and the output is the potential drop obtained across the load resistance R_L . The direction of flow of the current in the circuit is given by the arrows in the above diagram. The output waveform of the voltage is the same as that of the input.

During the negative half of the circuit the polarity gets reversed and hence the diode gets reverse biased and thus the diode allows no current to flow through it, that is, it blocks the current as it is a unidirectional device. Hence, no output is obtained at R_L and hence the waveforms only occur or appear at the positive half cycles and cuts the waveforms at the negative cycles. The input and output waveforms of a half rectifier is given below:





18.

3 Marks

(i) Which mode of propagation is used by shortwave broadcast services having frequency range from a few MHz, to 30 MHz? Explain diagrammatically how long distance communication can be achieved by this mode.

Ans: The radiowave directed towards the sky and reflected by the ionosphere is required location of the Earth is called a sky wave and these sky waves when reflected back can be received at any another point on Earth and this communication is known as sky wave communication or ionosphere propagation.

The outermost layer of the Earth is known as what is called as the ionosphere and the ionosphere is responsible for reflecting back the radio waves it encounters instead of absorbing them because it reflects back waves with short range frequencies only.

(ii) Why is there an upper limit to frequency of waves used in this mode?

Ans: The ionosphere tends to absorb the waves that are above 40MHz as waves with this frequency are said to penetrate through the ionosphere. These sky waves have a frequency lying in the range 2MHz to 20MHz and hence known as short wave band and hence are reflected back. This is the reason why this mode of propagation is restricted upto a few MHz and an upper limit to its frequency is set.

19.

3 Marks

(i) Identify the part of the electromagnetic spectrum which is:

(a) suitable for radar system used in aircraft navigation,

Ans: Microwaves

(b) produced by bombarding a metal target by high speed electrons.

Ans: X-rays

(ii) Why does a galvanometer show a momentary deflection at the time of charging or discharging a capacitor? Write the necessary expression to explain this observation

Ans: The deflection in the current will be due to the conduction current in the wire and the displacement current between the plates of the capacitor where the electrons start to flow from the positive to the negative plates during the charging cycle of the capacitor and hence current is produced and a momentary deflection is shown by the galvanometer when the capacitor gets fully charged.

Again during its discharging cycle the current is decreased and hence no deflection is observed in the galvanometer as the conduction and displacement current is zero.

The expression to explain this observation is given by the Maxwell's modified equation for conduction and displacement current. The displacement current was given by:

$$i_D = \epsilon_0 \frac{d\phi_E}{dt}$$

20. For a CE transistor amplifier, the audio signal voltage across the collector resistance of $2k\Omega$ is 2V. Suppose the current amplification factor of the transistor is 100, find the input signal voltage and the base current if the base resistance is $1k\Omega$. 3 Marks

Ans: The collector voltage and resistance as well as the base resistance of a common emitter transistor is given already.

Given,

$$R_c = 2k\Omega = 2 \times 10^3 \Omega$$

$$V_c = 2V$$

$$R_B = 1k\Omega = 1 \times 10^3 \Omega$$

The current amplification factor or the current gain of a CE transistor is defined as the ratio of the small change in the collector current and the base current since in a CE transistor the emitter region of the transistor is kept common. The current gain is already given to be 100. Hence, the current gain denoted by β is given as:

$$\beta = \frac{\Delta I_C}{\Delta I_B} \text{ -----(1)}$$

Since, the base current needs to be found the value of the collector current needs to be computed. By applying the ohm's law equation we get:

$$V_c = I_c R_c$$

By rearranging the terms we get:

$$I_c = \frac{V_c}{R_c}$$

Hence, by substituting the given values of collector resistance and voltage we get:

$$I_c = \frac{2}{2 \times 10^3} = 10^{-3} A$$

This value of collector current and the given current gain is substituted in equation (1) to get:

$$100 = \frac{10^{-3}}{I_B}$$

Hence, the base current will be:

$$I_B = \frac{10^{-3}}{100}$$

$$\Rightarrow I_B = 10^{-5} \text{ A}$$

This is the base current of the CE transistor.

21. Explain the following giving reasons:

3 Marks

(i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.

Ans: Monochromatic light is a source of light which has a single frequency throughout and hence when it hits the surface of the media of separation then it would be so that the vibration of the particles in the second media is same as the vibrations of the particles present in the incident light and thus the frequency remains constant and thus there would be no bending of light rays. Hence, the frequency remains unaffected because the refracted and reflected rays have the same frequency.

(iii) In the wave picture of light, the intensity of light is determined by the square of the amplitude of the wave. What determines the intensity in the photon picture of light?

Ans: The energy is not affected by the decrease in speed when light passes from rarer to denser medium because the energy of light only depends upon its frequency given by the quantum theory of light radiation. Also, if energy is to be decreased then there must be some loss of energy which contradicts the principle of law of conservation of energy.

(ii) When light travels from a rarer to denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave?

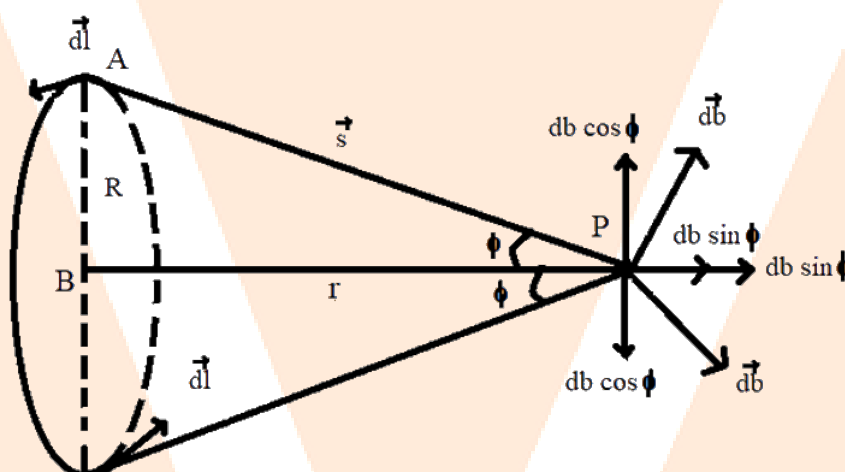
Ans: The intensity of light is governed by how many photons are incident or present per unit area in the surface of the material on which the incident light is incident on.

22. Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius R.

Draw the magnetic field lines due to a circular wire carrying current I.

3 Marks

Ans: The magnetic field is required to be found out at a point axial to the circular loop from the center of the circular loop. The diagram illustrating the same is given below:



Let $d\vec{l}$ be the current element over a small length dl where the direction of current is outwards and the position vector $\vec{s}$ indicates the distance between the loop and the axial point relative to the $d\vec{l}$ element. By applying the Biot-Savart's law we get:

$$dB = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{r}}{r^3}$$

Here, the distance considered is the vector $\vec{s}$ and hence the equation becomes:

$$dB = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{s}}{s^2}$$

$$\Rightarrow dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{s^2}$$

The angle between the current element $d\vec{l}$ and vector $\vec{s}$ is 90° because the circular loop is considered to be plane perpendicular to vector $\vec{s}$. Hence, we obtain:

$$\Rightarrow dB = \frac{Idl \sin 90^\circ}{s^2}$$

Since, $\sin 90^\circ = 1$ we have:

$$\Rightarrow dB = \frac{\mu_0 Idl}{4\pi s^2} \text{ -----(1)}$$

Let the angle made by the vector $\vec{s}$ with the line passing through the center of the circular loop be ϕ . Hence, there it can be resolved into its horizontal and vertical components indicated by the sin and cos components. The cos component and sin component represent the horizontal and vertical components respectively. This can be seen from the diagram above.

It can be seen that since there are two $\vec{s}$ from either side of the loop there will be two horizontal and vertical components each as shown in the above diagram. Here, since the sin components are in the same direction they get added up while the cos components cancel out each other as they are equal and opposite.

Hence, the total magnetic field at point P is given as:

$$\int dB = B = \int dB \sin \phi$$

From the triangle ABP the sin angle will be:

$$\sin \phi = \frac{R}{s}$$

This value is substituted back into the equation and also the value of the magnetic field from equation (1) is also substituted back to get:

$$B = \int \frac{\mu_0 Idl}{4\pi s^2} \times \frac{R}{s} = \int \frac{\mu_0 R Idl}{4\pi s^3}$$

By taking out the common terms we get:

$$B = \frac{\mu_0 RI}{4\pi s^3} \int dl$$

$$\Rightarrow B = \frac{\mu_0 RI}{4\pi s^3} \times L$$

Here, since L is the circumference of the circle we have $L = 2\pi R$ and hence the equation becomes:

$$B = \frac{\mu_0 R I}{4\pi s^3} \times 2\pi R$$

On further simplification we get:

$$B = \frac{\mu_0 R^2 I}{2s^3}$$

From the triangle ABP by applying the Pythagoras theorem we have:

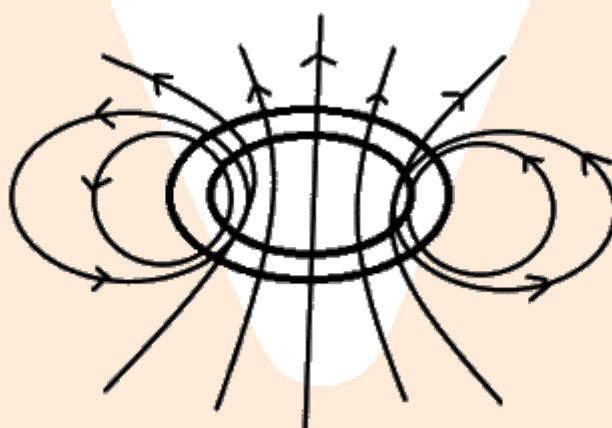
$$s = \sqrt{R^2 + r^2}$$

By substituting this in the equation for magnetic field we get:

$$B = \frac{\mu_0 R^2 I}{2(R^2 + r^2)^{\frac{3}{2}}}$$

Therefore this is the magnetic field of a point placed axially to a circular loop and the direction of magnetic field is along the positive X-direction.

The magnetic field lines due to a circular wire carrying current I is given by the diagram below:



PART-D

23. Ram is a student of class X in a village school His uncle gifted him a bicycle with a dynamo fitted in it. He was very excited to get it. While cycling during the night, he could light the bulb and see the objects on the road. He, however, did not know how this device works. He asked this question to his

teacher. The teacher considered it an opportunity to explain the working to the whole class. **4 Marks**

Answer the following questions:

(a) State the principle and working of a dynamo.

Ans: The dynamo machine is a generator which in this case is said to be attached to a cycle. The machine converts mechanical energy to electrical energy and as Ram cycles or pedals the current generated flows through the bulb and then the bulb is lit. The principle behind a dynamo is that the coil rotates in the presence of a magnetic field and cuts the lines of force to produce flux that is linked to the coil. This flux induces an emf and hence current is produced and this emf is induced due to the change in the flux that is linked to the coil in accordance to Faraday's law.

(b) Write two values each displayed by Ram and his school teacher.

Ans: The values shown by the student were enthusiasm, curiosity, determination, eagerness and an urge to gain knowledge and learn new concepts.

The values shown by the teacher was dedication, determination, generosity and enthusiasm to teach new concept to students and spread knowledge

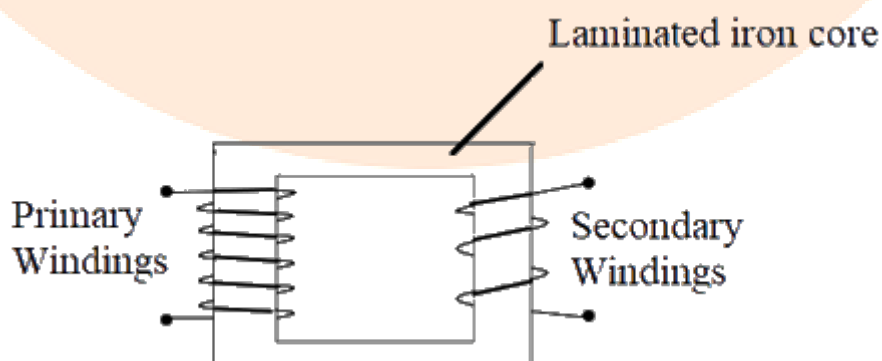
PART-E

24.

5 Marks

(i) Draw a labelled diagram of a step-down transformer. State the principle of its working.

Ans: The labelled diagram of a step-down transformer is given below:



A step-down transformer is an electrical device which converts an alternating current at high voltage into a low voltage, that is, it decreases the input voltage.

The principle behind a step-down transformer is that it works on the concept of mutual inductance wherein an alternating current passes through one of the two inductively coupled coils and generates an induced emf due to the flux linked in the coil.

(ii) Express the turn ratio in terms of voltages.

Ans: The turn ratio in terms of voltages of the primary and secondary coils is given as:

$$\frac{\varepsilon_2}{\varepsilon_1} = \frac{N_2}{N_1} \text{ -----(1)}$$

Where,

N_1 is the number of turns of the primary coil

N_2 is the number of turns of the secondary coil

ε_1 is the number of turns of the secondary coil

ε_2 is the number of turns of the secondary coil

(iii) Find the ratio of primary and secondary currents in terms of turn ratio in an ideal transformer.

Ans: An ideal transformer is to be considered which means that there will be no loss of energy and hence the output power will be equivalent to the input power and hence we have:

Output Power=Input Power

Since, $P=\varepsilon I$ we have:

$$\varepsilon_1 I_1 = \varepsilon_2 I_2 \text{ -----(2)}$$

Hence we obtain the relation:

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{I_2}{I_1}$$

From equation (1) we have:

$$\frac{\varepsilon_2}{\varepsilon_1} = \frac{N_2}{N_1}$$

Hence, the relation becomes:

$$\frac{I_2}{I_1} = \frac{\varepsilon_1}{\varepsilon_2} = \frac{N_2}{N_1}$$

Where,

I_1 is the current produced in the primary coil

I_2 is the current produced in the secondary coil

(iv) How much current is drawn by the primary of a transformer connected to 220 V supply when it delivers power to a 110 V - 550W refrigerator?

Ans: From the relation obtained from equation (2) we have:

$$\varepsilon_1 I_1 = \varepsilon_2 I_2$$

We consider an ideal transformer and the input power is already given to be 550W and also the input voltage that is given to the primary coil is given to be 220V as well. The current produced in the primary coil is required to be found. Hence the above obtained becomes:

$$P = \varepsilon_1 I_1$$

Hence the current in the primary coil after substituting the given values is:

$$I_1 = \frac{P}{\varepsilon_1}$$

$$\Rightarrow I_1 = \frac{550}{220} = \frac{5}{2}$$

$$\Rightarrow I_1 = 2.5A$$

Hence, this is the current in the primary coil.

Or

5 Marks

(a) Explain the meaning of the term mutual inductance. Consider two long concentric circular coils, one of radius r_1 and the other of radius r_2 ($r_1 < r_2$)

placed coaxially with centres coinciding with each other. Obtain the expression for mutual inductance of this arrangement.

Ans: Mutual inductance is the phenomenon of the generation of induced emf in one coil due to a change of current in the other neighbouring coil. There are two concentric circles with their radii given to be r_1 and r_2 where $r_1 < r_2$. Hence the magnetic field produced inside them will be due to the flux linked inside the coils and this flux is given as:

$$\phi = BA$$

Where, B is the magnetic field

A is the area of the coil

Mutual inductance is when the magnetic flux produced in one coil induces a flux in the other coil. Hence the flux equation becomes:

$$\phi_{12} = B_2 A_1 \text{ -----(1)}$$

Here, ϕ_{12} represents the flux produced inside coil 1 due to coil 2 and hence the magnetic field inside coil 1 will be that of the coil 2. Since, the coils considered here are circles, the area of coil 1 will be:

$$A_1 = \pi r_1^2$$

By substituting this in the above equation (1) we get:

$$\phi_{12} = B_2 (\pi r_1^2)$$

The magnetic field of coil 2 is in accordance to the Ampere's circuital law equation is given as:

$$B_2 = \mu_0 n_2 I_2$$

Here, n_2 represents the number of turns of the coil per unit length and hence:

$$n_2 = \frac{N_2}{l_2}$$

By substituting this in equation (2) we get:

$$B_2 = \mu_0 \left(\frac{N_2}{l_2} \right) I_2$$

The length of the coil here will be equivalent to the diameter of the circular coil and hence the length will be given as:

$$l_2 = 2r_2$$

Hence, by substituting this value of length back into the equation we get:

$$B_2 = \frac{\mu_0 N_2 I_2}{2r_2}$$

This is the magnetic field produced in coil 2 and hence substituting this back into equation (1) we get:

$$\phi_{12} = \frac{\mu_0 N_2 I_2}{2r_2} \times \pi r_1^2$$

From the basic formula for mutual inductance of two coils when magnetic flux is linked to one of them due to the other is given as:

$$\phi_{12} = M I_2$$

By equating both of these equations since they are similar, we get:

$$M I_2 = \frac{\mu_0 N_2 I_2}{2r_2} \times \pi r_1^2$$

The common term I_2 gets cancelled out and hence the mutual inductance value that is obtained is:

$$M = \frac{\mu_0 N_2 \pi r_1^2}{2r_2}$$

Hence, this is the expression for mutual inductance of two circular concentric coils.

(b) A rectangular coil of area A, having number of turns N is rotated at 'f' revolutions per second in a uniform magnetic field B, the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2 \pi f NBA$.

Ans: We consider the rectangular coil of an AC generator which rotates cuts the flux lines of a uniform magnetic field which leads to a generation of an induced emf. The angle of rotation that is the angle the coil makes with the magnetic field is at an angle θ then the total magnetic flux linked with the rotating coil will be:

$$\phi = BA \cos \theta$$

If the coil tends to rotate with an angular velocity ω through the angle θ in a time interval t then the angle θ will be:

$$\theta = \omega t$$

By substituting this in the equation for flux we get:

$$\phi = BA \cos(\omega t) \text{ -----(1)}$$

The induced emf that is produced in terms of the flux that is linked given in accordance to Faraday's laws is given as:

$$\varepsilon = -\frac{d\phi}{dt}$$

The value for flux from equation (1) is substituted in the above equation to get:

$$\varepsilon = -\frac{d}{dt}(BA \cos \omega t)$$

On simplifying further we get:

$$\varepsilon = -BA \frac{d}{dt}(\cos \omega t) = -BA\omega(-\sin \omega t)$$

$$\Rightarrow \varepsilon = BA \sin \omega t$$

Since, it is given that the coil has N number of turns, the above equation becomes:

$$\Rightarrow \varepsilon = NBA \omega \sin \omega t$$

We know that the angular frequency is:

$$\omega = 2\pi f$$

Hence, we substitute this in the emf equation to get:

$$\Rightarrow \varepsilon = NBA(2\pi f) \sin 2\pi ft$$

Here, f is already given to be the frequency of rotation of the coil. The induced emf is maximum only when the sin angle takes the value 90° , that is, when the plane of the coil is parallel to the magnetic field and hence by applying this we get:

$$\Rightarrow \epsilon = NBA(2\pi f)\sin 90^\circ$$

Since, $\sin 90^\circ = 1$ we have:

$$\epsilon_{\max} = 2\pi f NBA$$

Hence proved.

25.

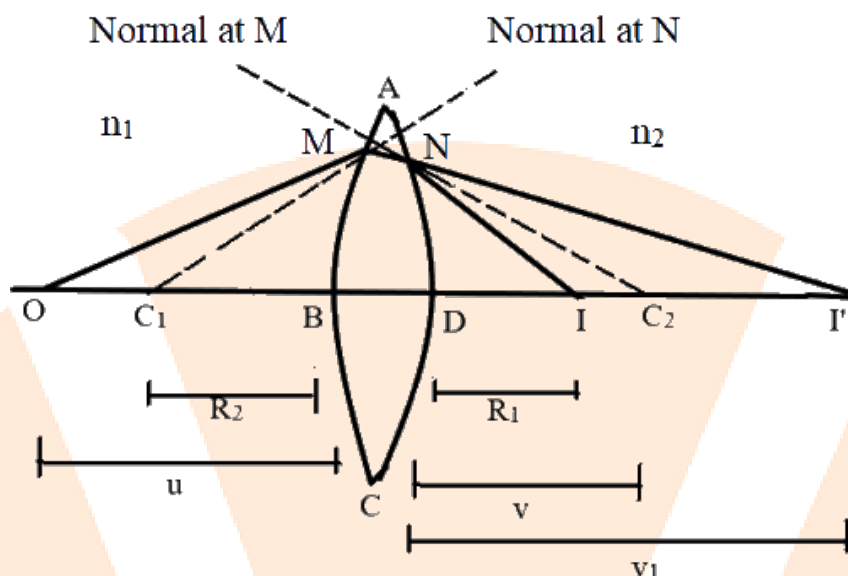
5 Marks

(i) Derive the mathematical relation between refractive indices n_1 and n_2 of two media and radius of curvature R for refraction at a convex spherical surface. Consider the object to be a point since lying on the principle axis in rarer medium of refractive index n_1 and a virtual image formed in the denser medium of refractive index n_2 . Hence, derive lens maker's formula.

Ans: There are certain assumptions that are to be considered for deriving the lens maker's formula. The assumptions that are taken are:

- The lens used is thin so that the distances measured from its surface may be taken equal to those measured from the optical centre.
- The aperture of the lens is considered small
- The object that is taken is a point object placed on the principle axis.
- All the rays are considered to be paraxial, that is, they make very small angles.

The diagram below illustrates the ray diagram to derive the lens maker's formula.



When we consider the refraction on aperture or the surface ABC only then the equation for spherical surfaces is applied to get:

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \text{ -----(1)}$$

The image after refraction at ABC surface is supposed to have formed an image at \$I'\$ given in the diagram but instead the rays suffer refraction again at surface ADC. Thus, when \$I'\$ is extended backwards, it acts as a virtual object for the refraction at surface ADC and hence the corresponding final real image that is formed is at I.

When we consider the refraction on the surface ADC only then the equation for spherical surfaces is applied to get:

$$\frac{n_1}{v} - \frac{n_2}{v_1} = \frac{n_1 - n_2}{R_2} \text{ -----(2)}$$

By adding up equations (1) and (2) we get:

$$\left[\frac{n_2}{v_1} - \frac{n_1}{u} \right] + \left[\frac{n_1}{v} - \frac{n_2}{v_1} \right] = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{R_2}$$

On solving further we have:

$$\frac{n_2}{v_1} - \frac{n_1}{u} + \frac{n_1}{v} - \frac{n_2}{v_1} = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{R_2}$$

$$\Rightarrow \frac{n_1}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{R_2}$$

The above equation can be written as:

$$\Rightarrow \frac{n_1}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} - \frac{n_2 - n_1}{R_2}$$

By taking n_1 on the left hand side and $(n_2 - n_1)$ on the right hand side we get:

$$\Rightarrow n_1 \left[\frac{1}{v} - \frac{1}{u} \right] = (n_2 - n_1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

By rearranging the terms we get:

$$\left[\frac{1}{v} - \frac{1}{u} \right] = \frac{(n_2 - n_1)}{n_1} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

We know from the thin lens formula that:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

By substituting this in the above equation we get:

$$\frac{1}{f} = \frac{(n_2 - n_1)}{n_1} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\Rightarrow \frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Thus, this is the expression for lens maker's formula.

(ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 and radius of curvature 20 cm. The distance of light source from the glass surface is 100 cm. At what position is the image formed?

Ans: Since, the light is incident on only one of the surfaces of the spherical convex lens the formula from equation (1) is applied. Hence, we get:

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$$

The refractive index of glass, the radius of curvature and the object distance is already given. The refractive index of air is already known to be 1 since the refraction occurs from air medium to glass medium. The image distance is to be found out.

By applying the sign conventions we get:

Given,

$$u = -100 \text{ cm}$$

$$R = 20 \text{ cm}$$

$$n_1 = 1$$

$$n_2 = 1.5$$

Hence by substituting these given values into the above equation we get:

$$\frac{1.5}{v_1} - \frac{1}{(-100)} = \frac{1.5 - 1}{20}$$

On simplifying further we get:

$$\frac{1.5}{v} + \frac{1}{100} = \frac{1.5 - 1}{20}$$

$$\Rightarrow \frac{1.5}{v} = \frac{0.5}{20} - \frac{1}{100}$$

$$\Rightarrow \frac{1.5}{v} = \frac{2.5 - 1}{100} = \frac{1.5}{100}$$

$$\Rightarrow \frac{1.5}{v} = \frac{1.5}{100}$$

By cross multiplying the terms we get:

$$\Rightarrow v = 100 \text{ cm}$$

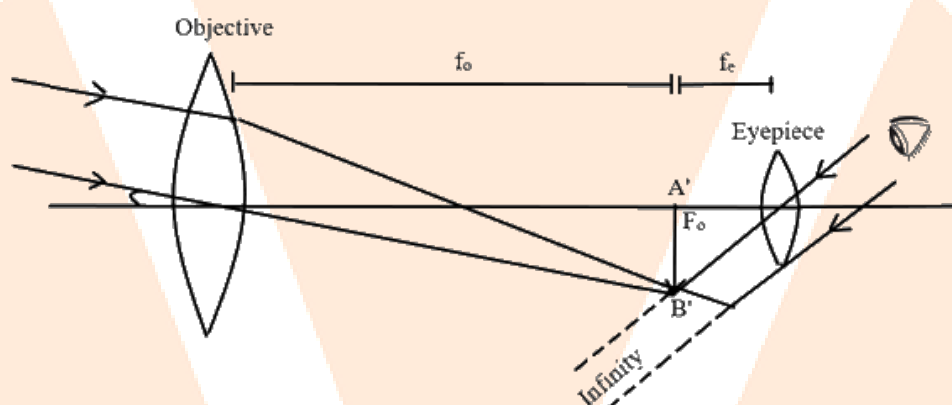
Hence the image is formed at a distance of 100 cm from the surface.

Or

5 Marks

a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.

Ans: The below diagram illustrates the ray diagram for an astronomical telescope in normal adjustment with its rays coming from infinity and the final image forming a real image. Thus, the diagram is shown below:



The magnifying power of the telescope in normal adjustment is said to be the ratio of the angle subtended at the eye by the final image as seen through the eyepiece of the telescope to the angle subtended at the eye by the object.

(b) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope.

(i) Which lenses should be used as objective and eyepiece? Justify your answer.

Ans: The lens of power 10 dioptre needs to be used as the lens for the eyepiece because this lens will be having a lower value of focal length since power and focal length are inversely related to each other and similarly for the objective the lens with power 0.5 dioptre needs to be selected.

This is done in order to produce a clearer and a more magnified image because magnifying power in terms of the focal lengths of the objective and eyepiece is given as:

$$m = \frac{f_o}{f_e} = \frac{P_e}{P_o}$$

(ii) Why is the aperture of the objective preferred to be large?

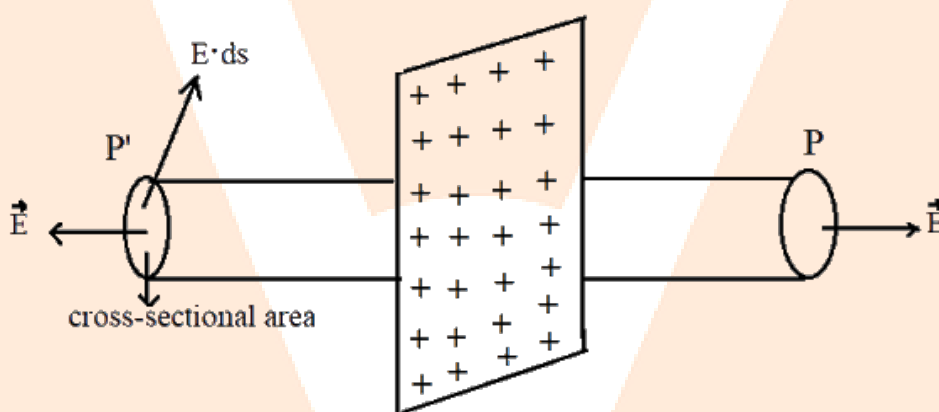
Ans: When the objective is made of a large aperture then it will be easier to absorb more light incident on it so that it can produce a more magnified image of the terrestrial and distant objects as its image.

26.

5 Marks

(i) Use Gauss's law to find the electric field due to a uniformly charged infinite plane sheet. What is the direction of field for positive and negative charge densities?

Ans: The diagram illustrated below is the electric field in a uniformly charged plane sheet:



The Gaussian surface which is an imaginary surface that is considered here is a cylindrical gaussian surface as shown in the diagram above is made in order to find out the total flux closed by the cylindrical Gaussian surface.

The electric field is set up to be set up due to the electrical charge present on the surface of the plane sheet

Electric field only exists in the cross-sectional area A and not in the curved surface area of the cylinder. It also has same magnitude and opposite directions given at points P and P'.

Since, in curved surface area, there is no flux that exists and hence we have:

$$\phi_E = 0$$

Hence, the flux at the ends, that is, the cross-sectional areas of the cylindrical Gaussian surface is considered. This flux is given as:

$$\phi_E = \vec{E} \cdot \vec{ds}$$

Since, there are two surfaces of cross-sectional area to be considered we have:

$$\phi_E = 2\vec{E} \cdot \vec{ds}$$

The dot product of two vectors are given as:

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

Hence, by applying this formula for electric field and area vector we get:

$$\phi_E = 2\vec{E} \cdot \vec{A} = 2EA \cos \theta \quad \text{-----(1)}$$

The surface charge density is defined as the charge on the surface per unit area and hence it is given as:

$$\sigma = \frac{q}{A}$$

By rearranging the terms we get:

$$q_{\text{net}} = \sigma A \quad \text{-----(2)}$$

From the Gauss' theorem we get the equation for net flux to be:

$$\phi_E = \frac{q_{\text{net}}}{\epsilon_0}$$

Hence by substituting equation (2) in the above equation we get:

$$\phi_E = \frac{\sigma A}{\epsilon_0}$$

By substituting the value of flux from equation (1) we have:

$$2EA \cos \theta = \frac{\sigma A}{\epsilon_0}$$

The angle between electric field and area vector is zero because they are parallel to each other and hence we have:

$$EA \cos(0^\circ) = \frac{\sigma}{2\epsilon_0}$$

Since, $\cos(0^\circ)$ we obtain the equation for electric field due to a uniformly plane sheet which is charged:

$$E = \frac{\sigma}{2\epsilon_0}$$

As per the principles of electrostatic forces, the electrical lines of force will be directed outwards for a sheet having positive charge densities and hence the electric field vector will be pointing in a direction away from the sheet.

As per the principles of electrostatic forces, the electrical lines of force will be directed inwards for a sheet having negative charge densities and hence the electric field vector will be pointing towards the sheet.

(ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1:2 so that the energy stored in the two cases becomes the same.

Ans: The energy or the potential energy stored by a capacitor in terms of capacitance and voltage is given as:

$$U = \frac{1}{2} CV^2$$

The energy stored in the case of both parallel and series connection is said to be equal and hence we have:

$$U_p = U_s$$

$$\Rightarrow \frac{1}{2} C_p V_p^2 = \frac{1}{2} C_s V_s^2$$

By cancelling out the common terms we get:

$$C_p V_p^2 = C_s V_s^2$$

From this equation the relation between capacitance ratios and voltage ratios can be determined and hence we have:

$$\frac{C_p}{C_s} = \frac{V_s^2}{V_p^2}$$

$$\Rightarrow \frac{V_s^2}{V_p^2} = \sqrt{\frac{C_p}{C_s}}$$

We know from the formula of resistances connected in series and parallel the equivalent resistance for resistors connected in series with each other is given as:

$$C_p = \frac{C_1 C_2}{C_1 + C_2}$$

We know from the formula of resistances connected in parallel the equivalent resistance for two capacitors connected in parallel with each other is given as:

$$C_s = C_1 + C_2$$

Hence, by substituting the respective values for parallel and series capacitance we get:

$$\frac{V_s}{V_p} = \sqrt{\frac{C_1 + C_2}{\frac{C_1 C_2}{C_1 + C_2}}}$$

On simplifying further we get:

$$\frac{V_s}{V_p} = \sqrt{\frac{(C_1 + C_2)^2}{C_1 C_2}} = \frac{C_1 + C_2}{\sqrt{C_1 C_2}}$$

$$\frac{V_s}{V_p} = \frac{C_1 + C_2}{\sqrt{C_1 C_2}} \text{-----(1)}$$

The ratio between the capacitor C_1 and C_2 parallel capacitance have been given as 1:2 and hence we can say that:

$$\frac{C_1}{C_2} = \frac{1}{2}$$

Thus we get:

$$C_2 = 2C_1 \text{-----(2)}$$

Hence by substituting equation (2) in equation (1) we get:

$$\frac{V_s}{V_p} = \frac{2C_1 + C_2}{\sqrt{2C_1 \times C_1}}$$

On simplifying further we get:

$$\frac{V_s}{V_p} = \frac{3C_1}{\sqrt{2C_1^2}} = \frac{3C_1}{\sqrt{2}C_1}$$

By cancelling out the common term C_1 we get:

$$\frac{V_s}{V_p} = \frac{3}{\sqrt{2}}$$

This is the ratio of the potential difference of the series and parallel connected capacitors.

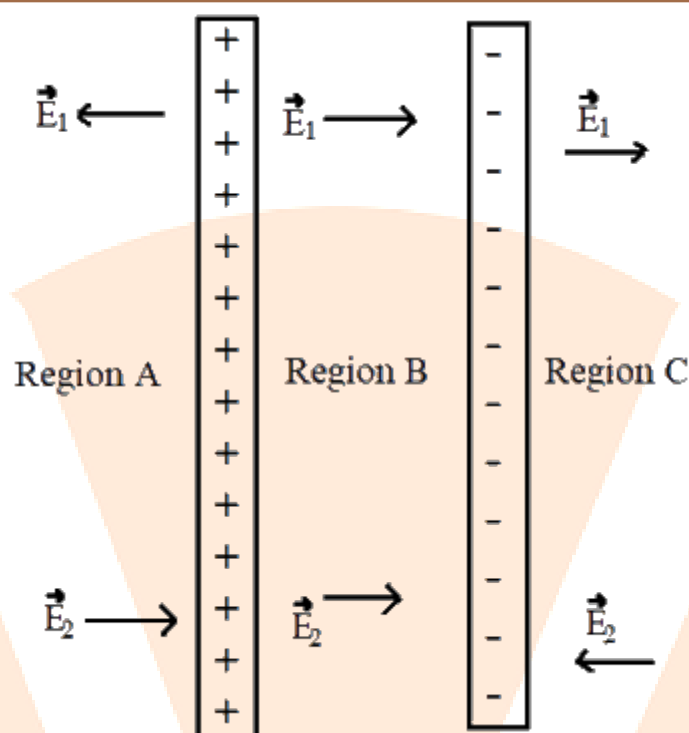
Or

5 Marks

(i) If two similar large plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$ are separated by a distance d in air, find the expressions for:

(a) field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.

(a) The diagram below illustrates two identical large plates with their respective surface charge densities which are placed parallel to each other separated by a distance d given already:



Due to the presence of charges on the surface of the sheets there will be an electric field that is induced. As seen, in the above diagram there are three regions to be considered. The first region that we consider is the region labelled A in the diagram.

In this region, the field lines or the lines of force will be directed outwards for the sheet 1 because the sheet is filled with positive charges and hence the electric field will be pointing outwards due to charge present in sheet 1.

Similarly, the electric field produced due to the charges present in sheet 1 in region A is also considered. The lines of force will be directed inwards due to the sheet 2 in the same region A and hence the electric field will be pointing inwards due to charges present in sheet 2.

The same principle applies for region C, where the electric field in region C will point outwards due to charges present in sheet 1 and the electric field due to sheet 2 will point inwards.

In the middle region B, the electric field due to sheet 1 will be directed outwards while the field due to sheet 2 will be directed inwards.

Let,

E_1 be the electric field produced due to sheet 1

E_2 be the electric field produced due to sheet 2

Hence the above stated directions of electric field vectors due to the charges present in both sheets is shown in the diagram below:

Hence, the net electric field in region A is:

$$E_{\text{net}} = E_1 - E_2$$

However the directions are clearly opposite to each other because the surface charge densities are $+\sigma$ and $-\sigma$ for the two sheets respectively. Thus, as per equation (1) we have:

$$E_{\text{net}} = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} = 0$$

Hence, due to the opposite directions of field vector the electric fields cancel out each other. Similarly, in region C also the net electric field is zero.

In region B, we have the electric field will be added up since they are in the same direction:

$$E_{\text{net}} = E_1 + E_2$$

$$\Rightarrow E_{\text{net}} = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$\Rightarrow E_{\text{net}} = \frac{\sigma}{\epsilon_0} \text{ -----(1)}$$

Hence, this is the net field due to these two sheets.

(b) the potential difference between the plates.

Ans: The potential in terms of the electric field, E , and the distance of separation between the two sheets, d is:

$$V = Ed$$

From equation (1) the electric field value is known, and hence by substituting in the above equation we get:

$$V = \frac{\sigma}{\epsilon_0} d$$

The surface charge density is defined as the charge per unit area and is hence given by:

$$\sigma = \frac{Q}{A}$$

By substituting in the above equation we get:

$$V = \frac{Qd}{A\epsilon_0} \text{-----(2)}$$

Hence, this is the potential difference between the sheets

(c) the capacitance of the capacitor so formed.

(c) the capacitance of the capacitor so formed.

Ans: The two long sheets with opposite polarity of charges on their surfaces act like a parallel plate capacitor.

The capacitance is given as:

$$C = \frac{Q}{V}$$

The value of potential difference obtained in equation (2) is substituted in the above equation to get:

$$C = \frac{Q}{\frac{Qd}{A\epsilon_0}} = \frac{A\epsilon_0}{d}$$

$$\Rightarrow C = \frac{A\epsilon_0}{d}$$

This is the capacitance of the capacitor so formed.

(ii) Two metallic spheres of radii R and $2R$ are charged so that both of these have same surface charge density σ . If they are connected to each other with a conducting wire, in which direction will the charge flow and why?

Ans: Due to the principle of conservation of charges the charge flows from the sphere with the high potential to the sphere with a lower potential but the total amount of charge is constant and remains unchanged. Thus, the charges are redistributed when they are connected by a conducting wire. As per the principle, charges flow from one conducting sphere to another until both of these spheres reach a common potential.

The surface charge density is given as:

$$Q = \sigma A \text{ -----(1)}$$

Hence, the conducting sphere which has a radius $4R$ would have more charge present on its surface than the sphere with radius R since it has a larger surface area comparatively because the surface area is directly proportional to the charge present on its surface as given equation (1).

The potential difference is given as:

$$V = \frac{Q}{C}$$

The potential is directly proportional to the charge hence the sphere given by equation (2) and thus the conducting sphere with a larger radius, that is, $4R$ will be at a higher potential and the flow of charges will hence be from sphere with radius $4R$ to sphere with radius R .